21/4/26

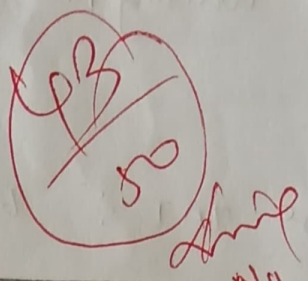
Test

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Course:- Operating System

Course code:- CSA0414

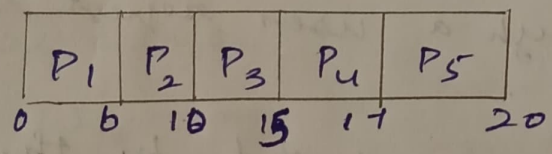


3) b)

Process	Arrival Time	Burst Time	Priority value
P1	0	6	3
P2	1	4	1
P3	2	5	4
P4	3	2	2
P5	4	3	5

Draw the Grant charts for FCFS :-

Solution:-



Process	Arrival Time	Burst Time	Priority value	Waiting Time	Turn around Time/TAT	CT completion time
P1	0	6	3	0	6	6
P2	1	4	1	5	9	10
P3	2	5	4	8	13	15
P4	3	2	2	12	14	17
P5	4	3	5	13	16	20

i) Average waiting Time :- $\frac{0+5+8+12+13}{5}$

$= 7.6$

ii) Averge TAT : $\frac{6+9+13+14+16}{5}$

$= 11.6$

a) System call is a mechanism thorough which a user program request using a service in operating system :-

Ty: Types of system calls :-

*) Process Control
*) File Management
*) Service Management
*) Information management.

How the system call works :-

*) It works through a user request to the operating system.

*) So, the service is controlled by the Process.

*) System call is requested.
*) Control move to kernel
*) Kernel execute the request.
*) Request return to the user.

1) Evolution of operating system:-

*) Operating system is an software system. Interacts between users interface and the computer hardware.

*) It mainly supports CPU utilization, system sharing, memory, files and resources.

*) There are mainly four types of operating system:-

-) Modern operating system
-) Branch operating system
-) Multi programming operating system.
-) Time sharing operating system.

4
*) Branch operating system:-

-) Similar jobs are connected and they are grouped together as separate branches.
-) Jobs are performed without any direct user interface.
-) If any job requires or waits for I/o at that time CPU remains idle.
-) Eg:- IBM mainframe system.

2) Multiprogramming operating system:-

*) It allows multiple program to run simultaneously.

*) If one program waits for Input and output then the CPU process the other program to run.

*) This improves CPU efficiency.

*) In the next process CPU decide which Program should run next.

*) This improves overall CPU runs efficiently.

*) Eg:- Early IBM operating system

3) Modern operating system:-

*) Modern operating system includes multitasking, multi programming and response time.

*) It provides file management, memory management and security.

*) They also help in cloud computing and mobile computing.

*) Eg:- Unix, macos, android etc.)

*) Time-sharing operating system

*) It allows multiple users to use the computer system efficiently.

*) It has various time intervals or breaks known as time-slice or time quantum.

*) Eg:- Linux operating system.

4)

4)

Process	Allocation A	B	C	Max A	B	C	Need			Available		
P0	0	1	0	7	5	3	7	4	3	7	2	6
P1	2	0	0	3	2	2	1	2	2	9	2	6
P2	3	0	3	9	0	2	6	0	1	12	2	9
P3	2	1	1	4	2	2	2	1	1	14	3	10
P4	0	0	2	5	3	3	5	3	1	14	3	12
	7	2	6									

Formula:-

Need = Max - Allocation.

if Need <= work , then work + Allocation

P0 :- 7432 = 726 (It is true)

P0:- (waiting) -> waiting

Available :- 726 + 010 = 736

P1 :- 122 <= 736 (It is true)

Available = 726 + 200
= 926

P2 :- 601 <= 926 (It is true)

Available = 926 + 303
= 1229

2) Intro:-

Multiprocessing is a technique in which a computer system uses two or more Processors (CPUs) to execute multiple tasks simultaneously. It improves system performance, speed, reliability.

There are two types of multiprocessing:-

1) Symmetric Multiprocessing (SMP):-

*) In SMP, all Processors are equal and share the same memory and operating system.

*) Each Processor can perform any task.

*) The OS schedules processes among all available Processors.

*) If one processor is busy, another Processor can execute a different task.

eg:- A computer with four CPU cores where all cores execute different applications simultaneously.

2) Asymmetric Multiprocessing (AMP):-

*) In AMP, Processors are assigned different responsibilities.

*) One processor acts as the master and control the system.

*) The master processor handles scheduling and distributes work to other processors.

Eg:- In an embedded system, one Processor controls the system while other Processors Perform specific operations.

Difference between SMP and AMP

Symmetric Multiprocessing	Assymmetric Multiprocessing
*) All processors are equal	*) Processors having different roles.
*) Each Processor runs any task.	*) Each processor gets specific tasks.
*) Better load balancing	*) Work is assigned by the master.

Advantages:-

*) Increased Throughput:
*) Time Saving
*) Economy of Scale.
*) Increased Reliability.
*) Better performance.

Conclusion:-

Multiprocessing improves computer performance by allowing multiple Processors to execute tasks simultaneously.

2) a) Introduction:-

*) A system call is a mechanism through which a user Program requests a service from the operating system kernel.

*) It acts as an interface between user Applications and the operating systems.

How a system call is executed :-

User Program (Requests an OS service)

↓

System call interface
(identifies the requested system call)

↓

Kernel mode
(OS executes the requested service)

↓

Return Result
(Return is returned to the User program)

The categories of system calls :-

1) Process control
2) File Management
3) Device Management
4) Information Maintenance
5) Communication.

Conclusion :-

System calls provide communication between user programs and the OS kernel and allow applications to access system resources securely.

b) Single-Threaded Process :-

*) The server creates a separate child process for each client request

*) Each process handles one request independently.

*) Processes have separate memory spaces.

*) Process creation and context switching require more resources.

eg:- When 1000 clients send requests, the server may create 1000 child processes.

b) Multithreaded process :-

*) A single server process creates a new thread for each client request.

*) threads share the same process memory and resources.

*) Each thread handles a separate client request.

*) Thread creation is faster and requires fewer resources than process creation.

Eg:- When 1000 clients send requests, the server creates 1000 threads to handle them.